MECH 191 — Metallurgy

Week 15

Weld Metallurgy & Failure

November 30, 2026 · 5:00 – 7:30 PM · Kai Santos

Class Agenda

01

Weld Metallurgy

Weld metal microstructure, HAZ sub-zones, and what the thermal cycle does to each region

02

Weld Cracking

Hot cracking, hydrogen-induced cold cracking, and lamellar tearing — mechanisms, conditions, and prevention

03

Fracture & Failure Analysis

Ductile, brittle, and fatigue fracture surfaces — and how a technician investigates a failed component

Metallography Lab — Module 14

Module 14 · Week 15

Weld Metallurgy & Failure

What to do this week:

Measure CGHAZ grain size vs. base metal at 100×. Identify ductile vs. brittle fracture zones in N.

Steps:

1 Observe — complete feature table
2 Sketch — label magnification & specimen
3 Answer — 4 short questions

Specimens to Check Out

· Specimen J — weld cross-section

· Specimen N — fracture cross-section

How to check out the kit:

1 See instructor after class today
2 Sign out the specimen tray
3 Use the EXAMET-5 at your convenience
4 Return kit before leaving class
5 Submit completed module with your portfolio

MECH 191 — Course & Unit Roadmap

Unit 1 Materials Fundamentals	Unit 2 Steel & Alloys	Unit 3 Testing & Quality	Unit 4 Manufacturing	Unit 5 Joining & Failure
Wk 1 Aug 24 Atomic Structure & Bonding	**Wk 4** Sep 14 Carbon & Alloy Steels	**Wk 7** Oct 5 Mechanical Testing	**Wk 10** Oct 26 Casting	**Wk 14** Nov 23 Welding Processes
Wk 2 Aug 31 Mechanical Properties	**Wk 5** Sep 21 Stainless & Cast Irons	**Wk 8** Oct 12 NDT Methods	**Wk 11** Nov 2 Forming	**Wk 15 ← here** Nov 30 Weld Metallurgy & Failure
	Wk 6 Sep 28 Nonferrous & Phase Diagrams	**Wk 9** Oct 19 Quality Standards	**Wk 12** Nov 9 Machining	**Wk 16** Dec 7 Corrosion & Wrap-up
			Wk 13 Nov 16 Powder Metallurgy	

Week 17 • FINAL EXAM

Week 15 — At a Glance

Topics

Weld metal microstructure — columnar solidification, acicular ferrite, allotriomorphic ferrite, bainite, and martensite
HAZ sub-zones — CGHAZ, FGHAZ, ICHAZ, SCHAZ: peak temperatures, microstructures, and property implications
Hot cracking — solidification cracking and crater cracks: low-melting films, Mn:S ratio, prevention
Hydrogen-induced cracking — three conditions, delayed nature, preheat, low-hydrogen consumables, inspection timing
Lamellar tearing — through-thickness failure, MnS inclusions, Z-quality steel, UT detection
Fracture types — ductile (dimples, cup-cone), brittle (chevrons, river marks), fatigue (beach marks, striations)
Failure analysis — preserve, document, examine: the technician's role in root cause investigation

Resources

Reference Material

Moniz Ch. 27 & Ch. 8 · Kalpakjian Ch. 32
Weld Metallurgy & Failure Analysis

Visualizers

Weld Microstructure & HAZ (Interactive)
Fracture Types & Surface Reading
Weld Cracking — Hot, Cold & Lamellar
SEM Fractography (Interactive)

Key Reference

ESAB Welding Metallurgy Handbook
ASM Vol. 11 — Failure Analysis
Lincoln Electric HIC & Preheat Guide

Next: Week 16 — Corrosion & Wrap-up

Core Questions

Try to answer these before we dive in — we'll revisit them with answers at the end of class.

1. What are the metallurgical zones in a weld and what microstructure does each contain?
2. What is hydrogen-induced cracking and why is it the most dangerous welding defect?
3. What are the main fracture types and how do you identify them from the fracture surface?
4. How does a technician approach a failure analysis investigation?

Weld Zones & Microstructure

Metallurgical Zones in Every Fusion Weld

Weld Metal

Fully molten region — solidifies as cast metal with columnar grains growing from the fusion boundary toward the weld centerline. Composition = filler metal + diluted base metal. Target microstructure in low-alloy steel: acicular ferrite, for its combination of strength and toughness.

Heat-Affected Zone (HAZ)

Base metal heated above ~730°C but not melted — enough to alter its microstructure. Subdivided into sub-zones by peak temperature. The most metallurgically complex and most failure-prone region of any welded joint.

Unaffected Base Metal

Beyond the HAZ — original microstructure fully preserved. Properties match the mill certificate. Serves as the reference baseline against which HAZ and weld metal properties are compared.

Weld Metal Microstructure — Low-Alloy Steel

Solidification is epitaxial — grains nucleate at the fusion boundary and continue the crystal orientation of the partially melted base metal grains. Columnar grains grow perpendicular to the fusion boundary toward the weld centerline, producing anisotropic properties.

Acicular ferrite

Fine, interlocking ferrite plates nucleating on non-metallic inclusions — the most desirable microstructure. Deflects crack propagation and provides excellent strength-toughness balance. Promoted by controlled oxygen content and inclusion chemistry.

Allotriomorphic ferrite

Forms at prior austenite grain boundaries on cooling. Provides easy crack propagation paths along boundaries — reduces toughness. Minimized by composition and cooling rate control.

Bainite / Martensite

Form at higher cooling rates or in higher CE weld metals. Increase strength but reduce toughness. Martensite in weld metal is generally undesirable and requires tempering via PWHT.

Dilution by process: SMAW 20–30% · GMAW 25–35% · SAW 50–70%
The WPS specifies filler metal to account for dilution and achieve required weld metal properties.

Heat-Affected Zone — Sub-Zones & Critical Properties

HAZ Sub-Zones by Peak Temperature

CGHAZ >1100°C

Coarse-Grained HAZ — extensive austenite grain growth (5–10× base metal). Cools to coarse upper bainite or martensite. Lowest toughness of any weld zone. Primary site for hydrogen-induced cracking.

FGHAZ 900–1100°C

Fine-Grained HAZ — fully austenitised but insufficient time for grain growth. Cools to a fine-grained microstructure. Typically the toughest region in the entire weld.

ICHAZ 730–900°C

Intercritical HAZ — partially austenitised (between Ac1 and Ac3). Mixed microstructure of transformed and untransformed regions. Can be a site for localized softening in HSLA steels.

SCHAZ <730°C

Subcritical HAZ — no phase transformation. Tempering and stress relief only. Properties approach base metal. Typically no toughness concern.

HAZ Width, Multi-Pass Welds & CGHAZ Risk

HAZ width

Determined by heat input and base metal thermal conductivity. Higher heat input → wider HAZ. Aluminum has higher thermal conductivity than steel → produces a proportionally wider HAZ for the same heat input.

Multi-pass welds

Each subsequent pass reheats the HAZ of previous passes. The reheated CGHAZ (now an ICHAZ of the prior pass) can be partially retransformed, refining the coarse grain structure. Well-designed multi-pass procedures exploit this to improve HAZ toughness.

Why CGHAZ is critical

The CGHAZ combines the worst of all factors: coarse grains reduce toughness; martensite/bainite raise hardness; weld toe is a stress concentration; and this is where hydrogen concentrates during cooling. All three HIC preconditions converge here.

Weld Cracking — Hot Cracking & Lamellar Tearing

Hot Cracking — Solidification Cracking

Mechanism

Liquid films at austenite grain boundaries cannot withstand tensile stress from weld shrinkage during solidification. Most common in alloys with wide solidification temperature ranges and high S or P content — these elements concentrate in the last liquid to solidify and form low-melting films.

Appearance

Centerline cracks along the weld bead (solidification cracking) or open craters at run ends (crater cracks). Crack surfaces are oxidized from exposure to the arc environment at high temperature — a key distinction from cold cracking.

Prevention

Control S and P in base metal and filler; maintain Mn:S ratio >25:1 (Mn ties up S as MnS with a higher melting point); use low crack-susceptibility filler metals; fill craters properly at run ends; reduce restraint.

Detection

VT and PT for surface-breaking cracks; RT or UT for internal centerline cracks that do not break the surface.

Lamellar Tearing

What it is

A base metal failure unique to welded joints — occurs in the through-thickness direction of rolled plate under the transverse shrinkage stress of a fillet or T-joint weld. It is not a weld defect — the weld itself may be perfectly sound.

Cause

Elongated MnS and silicate inclusions aligned parallel to the plate rolling surface provide easy fracture paths under through-thickness tensile stress. Rolled plate has much lower ductility in the through-thickness direction than in the rolling plane.

Appearance

A stepped, terraced crack below and parallel to the fusion line, in the base metal rather than the HAZ. The characteristic stepped morphology (tread-and-riser pattern) is diagnostic — cracks run along inclusion planes and connect via short through-thickness steps.

Prevention & Detection

Use low-sulfur steel (S <0.005%) with good through-thickness ductility (Z-quality plate, tested by through-thickness tensile test — Z25 or Z35 grade). Redesign joints to reduce through-thickness stress; use buttering passes. Detection: UT is primary — characteristic subsurface location and morphology.

Hydrogen-Induced Cracking (HIC) — Cold Cracking

HIC is uniquely dangerous because it is delayed — a weld that looks and tests sound immediately after completion can develop cracks hours to days later.

1 Susceptible Microstructure

Hard martensite or coarse bainite in the CGHAZ — formed when hardenable steel (high CE) cools rapidly. The harder the microstructure, the more susceptible to HIC. Controlled by preheat (slows cooling) and keeping CE below critical thresholds.

2 Hydrogen Source

Atomic hydrogen generated in the welding arc dissolves into the weld pool. Sources: moisture in consumable flux/coating, base metal surface contamination, humid atmosphere. E7018 low-hydrogen electrodes must be stored in a heated oven; use within 4 hours of removal.

3 Tensile Stress

Weld shrinkage stress and external restraint drive hydrogen toward areas of high stress concentration — typically the weld toe in the CGHAZ. Higher restraint = greater risk. Minimize by joint design, welding sequence, and intermediate stress relief.

Mechanism: H diffuses to hard microstructure at high-stress sites → reduces cohesive strength → HIC initiates and grows. **Prevention:** Preheat + low-H consumables + minimize restraint — all three must be controlled simultaneously. **Detection timing:** Inspect ≥24–48 h after welding on hardenable steels per AWS D1.1.

Fracture Types — Ductile & Brittle

Ductile Fracture

Mechanism

Microvoid coalescence (MVC) — voids nucleate at inclusions and second-phase particles, grow under applied stress, and coalesce to form the fracture surface.

Fracture surface

Dull, fibrous, gray appearance. Dimpled texture under SEM — each dimple represents one coalesced void. Rough to the touch.

Macroscopic signs

Significant plastic deformation — necking in round specimens, distortion of flat sections. Cup-and-cone profile in round bars (flat central zone + 45° shear lip). The two halves do not fit back together cleanly.

Conditions that promote

High temperature, low strain rate, low strength, high toughness, small section size (lower triaxiality). Most common failure mode in ductile metals at ambient temperature under static overload.

Brittle Fracture

Mechanism

Cleavage — atoms separate along specific crystallographic planes with minimal energy absorption. Grain boundaries may arrest cleavage cracks, causing short intergranular segments between cleavage facets.

Fracture surface

Flat, bright, granular, crystalline appearance. Chevron marks (V-shaped ridges) point back toward the fracture origin. River marks (fine lines on cleavage facets) show crack propagation direction within each grain.

Macroscopic signs

No visible plastic deformation — the two fracture halves fit back together. Fracture surface is perpendicular to the maximum tensile stress. Failure is sudden and catastrophic with no warning.

Conditions that promote

Low temperature (below DBTT — ductile-to-brittle transition temperature), high strain rate (impact loading), triaxial stress state (thick sections, notches), high-strength/low-toughness microstructure.

Fatigue Fracture — Reading the Fracture Surface

Initiation & Slow Crack Growth Zone

Initiation requires a stress concentration at or near the surface. Common sites in welded components: weld toe, undercut, machining marks, corrosion pits, or near-surface inclusions.

Beach marks (arrest marks)

Curved lines showing successive positions of the crack front — visible to the naked eye. Form during crack arrest periods (rest, low load). Spacing increases as the crack grows and stress intensity rises. The beach mark zone is the entire slow crack growth region.

Striations

Microscopic ridges — one formed per load cycle, visible only under SEM. Spacing equals crack growth per cycle and can be used to calculate how long cracking was occurring. Parallel striations on a flat zone are the SEM signature of fatigue.

Ratchet marks

Steps between multiple initiation sites — indicate high stress or multiple stress concentrations. Each step is where two independent crack fronts have joined. More ratchet marks = more severe stress concentration.

Overload Zone & Reading the Fracture

Final overload zone

Rough fast fracture at the opposite edge from initiation. Its proportion of the total fracture area indicates applied stress: small fast zone = low stress, many cycles; large fast zone = high stress, few cycles.

Reading the full story

Start at the initiation site (apex of the beach mark arc at the surface). Follow beach marks outward as spacing increases. Locate the overload zone — its size indicates stress level. Count ratchet marks for concentration severity.

Fatigue in weld toes

The weld toe is the most common initiation site — it combines geometric stress concentration with tensile residual stress. Post-weld toe grinding, TIG dressing, or peening remove the concentration and introduce compressive residual stress to extend fatigue life.

Distinguishing from other modes

Fatigue: beach marks, smooth, surface-initiated. Brittle: chevrons, flat, crystalline, no deformation. Ductile: dimpled, gray, plastic deformation. Fracture mode identification is the first and most time-critical step of any investigation.

Technician's Perspective

Weld metallurgy and fracture mechanics only matter when someone has to act on them — here is what that looks like in practice.

Red Flag: HIC found 72 hours after SMAW on high-CE structural steel — weld passed VT at completion

A welder completes a highly restrained structural connection on A514 steel (CE ≈ 0.55). Preheat was applied and verified at the start of the joint, but was not maintained throughout a long multi-run sequence — the joint cooled between runs on a cool morning. VT immediately after welding shows no defects. 72 hours later, MT reveals tight transverse cracks at the weld toes in the CGHAZ. Action: for high-CE steels, preheat is an interpass requirement, not just a starting condition; maintain throughout the entire joint. Per AWS D1.1, final MT inspection must not be performed until at least 48 hours after completion of welding on any joint in a high-CE steel.

Caution: Fatigue fracture at a crane boom weld toe — the weld meets the drawing, but the joint still fails

A crane boom fails in service after 3 years. The fracture surface shows clear beach marks initiating at a weld toe with a sharp profile and visible undercut. The weld meets all dimensional requirements on the drawing. Root cause: the weld toe is a stress concentration — even a code-compliant weld toe creates a fatigue initiation site under cyclic loading. No post-weld toe treatment was specified. Action: fatigue-critical welds require more than dimensional conformance; specify toe angle, post-weld toe grinding or TIG dressing, and verify surface condition. Raise an NCR and an engineering review for the remaining joints.

Good Practice: Preserving the fracture surface — the difference between a defensible investigation and a dead end

A technician receives a failed shaft removed from service. Before doing anything else, they photograph the component in its as-found condition, bag each fracture half separately in clean plastic, label with part number, date, and location found, and store in a dry environment. The fracture surface later reveals fatigue striations under SEM with an initiation site at a machining mark. Preservation allowed the fractographic evidence to survive for definitive laboratory analysis. Action: never clean, separate without care, or allow corrosion to develop on a fracture surface. The fracture is the primary evidence — treat it accordingly from the moment of recovery.

Core Question Answers

What are the metallurgical zones in a weld and what microstructure does each contain?

Three zones: Weld Metal (fully molten, solidifies to columnar grains; acicular ferrite is the target microstructure for toughness). HAZ (not melted but heat-altered, subdivided into CGHAZ >1100°C with coarse grains/martensite/lowest toughness; FGHAZ 900-1100°C with fine grains/best toughness; ICHAZ partially transformed; SCHAZ tempered only). Unaffected base metal beyond the HAZ retains original microstructure.

What is hydrogen-induced cracking and why is it the most dangerous welding defect?

HIC requires three simultaneous conditions: susceptible microstructure (martensite/bainite in CGHAZ), hydrogen (from consumables/moisture/atmosphere), and tensile stress (weld shrinkage/restraint). Hydrogen dissolves in the arc and diffuses to high-stress hard microstructure regions, reducing cohesion. It is uniquely dangerous because it is delayed — cracks develop hours to days after a weld passes VT. Prevention: preheat + low-hydrogen consumables + minimize restraint. Inspection timing: ≥48 h after welding on hardenable steels.

What are the main fracture types and how do you identify them from the fracture surface?

Ductile: dull, fibrous, gray, dimpled (SEM); cup-and-cone in round bars; significant deformation; halves don't fit back. Brittle: flat, bright, granular, crystalline; chevron marks pointing to origin; river marks; no deformation; halves fit back; sudden failure. Fatigue: smooth beach mark zone (slow growth, visible to naked eye) + rough fast fracture zone; striations under SEM; initiation at a surface stress concentration; ratchet marks if multiple initiation sites.

How does a technician approach a failure analysis investigation?

(1) Preserve — secure immediately; do not clean, separate carelessly, or allow further damage. (2) Document — photograph in as-found condition; record service history, loading, environment, maintenance. (3) Visual examination — identify fracture type, locate initiation site, note crack path direction. (4) Non-destructive examination — MT/PT for additional cracks; UT for subsurface; dimensional check. (5) Mechanical and metallographic testing — hardness survey, cross-section metallography, SEM fractography. (6) Root cause and report — classify failure mode; identify root cause; recommend corrective action.

Key Takeaways — Week 15: Weld Metallurgy & Failure

1. The HAZ is the most metallurgically complex and vulnerable region of any welded joint — understanding what happens to each sub-zone during the thermal cycle explains why preheat, heat input control, and PWHT are specified, and why hydrogen cracking almost always initiates in the coarse-grained HAZ rather than in the weld metal or the base metal.

2. Hydrogen-induced cracking is uniquely dangerous because it is delayed — a weld that looks and tests sound immediately after completion can develop cracks hours or days later; for hardenable steels, inspection timing is a code requirement, not a convenience, and a VT performed immediately after welding does not satisfy the requirement for a 48-hour hold.

3. The fracture surface is a record of how a component failed — ductile dimples, cleavage facets, beach marks, and striations each tell a specific story; a technician who can read a fracture surface can identify the failure mode, locate the initiation site, and direct the investigation toward the root cause before any laboratory testing is done.

Resources & What's Next

Week 15 Resources

Moniz Ch. 27 & Ch. 8 · Kalpakjian Ch. 32
Welding Metallurgy and Failure Analysis — primary reading

Weld Microstructure & HAZ Visualizer
Interactive — zone boundaries, sub-zone microstructures, multi-pass effects

Fracture Types & Surface Reading
Interactive — ductile, brittle, and fatigue surface feature identification

Weld Cracking Types (Hot, Cold & Lamellar)
Interactive — mechanism, morphology, and prevention for each type

ESAB Welding Metallurgy Handbook
Primary reference for weld metal and HAZ microstructure

ASM Vol. 11 — Failure Analysis & Prevention
Standard reference for failure investigation methodology

Week 16 — Corrosion & Wrap-up

December 7, 2026 · Unit 5 concludes

- Electrochemical corrosion — the four-element cell and how to break it
- Uniform, galvanic, pitting, crevice corrosion, and SCC
- Protection methods — coatings, cathodic protection, inhibitors
- Material selection and design for corrosion resistance
- Course wrap-up — key themes across all 5 units